

Experiment -1

Linux Programming Environment & Basic Commands

A) Setting Up a Linux Programming Environment

C programming can be done **online or offline**. For this course, the programs should be compiled and executed in a **Linux environment**, since Linux provides standard C development tools such as GCC and a command-line terminal.

1. Online Method – CoCalc

CoCalc is a browser-based computing platform that provides access to a Linux terminal without requiring Linux to be installed on the computer.

Steps:

1. Open the CoCalc website.
2. Create an account or sign in.
3. Create a **new project**.
4. Open a **Linux Terminal** inside the project.
5. Create a C source file, for example:

```
nano hello.c
```

6. Write and save the C program.
7. Compile it using GCC:

```
gcc hello.c -o hello
```

8. Execute the program:

```
./hello
```

Advantage: No Linux installation is required; only a browser and internet connection are needed.

2. Offline Methods

Linux can also be set up locally using several methods.

a) Dual Boot

In **dual boot**, two operating systems, such as Windows and Linux, are installed on the same computer.

When the computer starts, the user chooses which operating system to run.

Advantages: Linux gets direct access to the hardware and generally provides good performance.

Disadvantages: Disk partitioning is required, and switching operating systems requires restarting the computer.

b) Single Boot

In **single boot**, Linux is installed as the main or only operating system on the computer.

Advantages: Full system resources are available to Linux and performance is generally excellent.

Disadvantage: The existing operating system may need to be removed or replaced.

c) Bootable Pendrive / USB

A Linux ISO image can be written to a USB drive to create a **bootable USB**.

The computer can then boot Linux directly from the USB. Many Linux distributions provide a **Live Linux** mode that can be used without installing Linux on the hard disk.

Advantages: Portable and useful for testing Linux without permanent installation.

Disadvantages: It may be slower than an installed operating system, and changes may not be saved unless persistent storage is configured.

d) Virtual Machine (VM)

A **Virtual Machine** allows one operating system to run inside another operating system.

For example:

Windows Host → Oracle VirtualBox → Ubuntu Linux VM

The physical computer is called the **host machine**, while the operating system running inside the VM is called the **guest operating system**.

This is a convenient method for learning Linux because Windows and Linux can be used simultaneously.

Virtual Machine Manager (Hypervisor)

A **hypervisor** is software or firmware that creates, runs, and manages virtual machines.

Hypervisors are mainly classified into two types.

Type-1 Hypervisor – Bare-Metal

A **Type-1 hypervisor** runs **directly on the physical hardware** rather than on top of another operating system.

Structure:

Hardware

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Type-1 Hypervisor

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Virtual Machines

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Guest Operating Systems

Examples:

- VMware ESXi
- Microsoft Hyper-V
- Xen

Type-1 hypervisors are commonly used in **servers, data centers and enterprise environments** because they provide high performance and efficient resource management.

Type-2 Hypervisor – Hosted

A **Type-2 hypervisor** runs as an application on top of an existing **host operating system**.

Structure:

Hardware

↓

Host Operating System

↓

Type-2 Hypervisor

↓

Virtual Machine

↓

Guest Operating System

Examples:

- **Oracle VirtualBox**
- VMware Workstation

Type-2 hypervisors are convenient for **students, developers and personal computers** because they are relatively easy to install and use.

Oracle VirtualBox

For this experiment, **Oracle VirtualBox** can be used to create the Linux programming environment.

The basic setup is:

Physical Computer

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Windows / Host OS

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Oracle VirtualBox

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Linux Virtual Machine

↓

Terminal + GCC

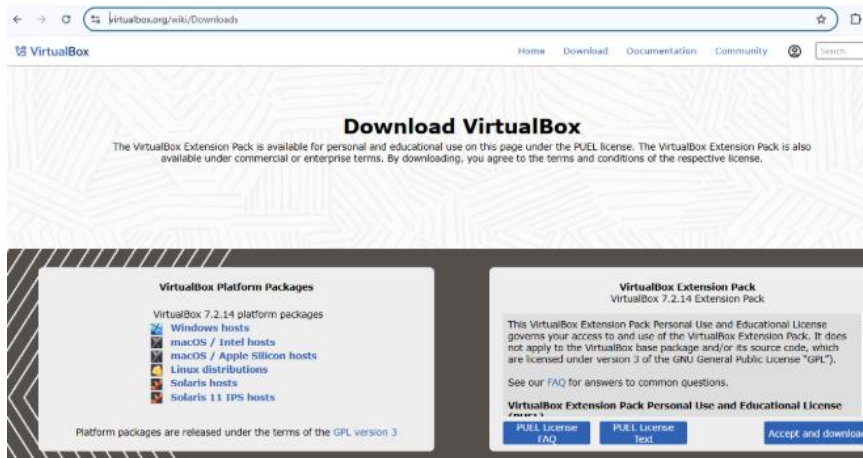
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C Programming

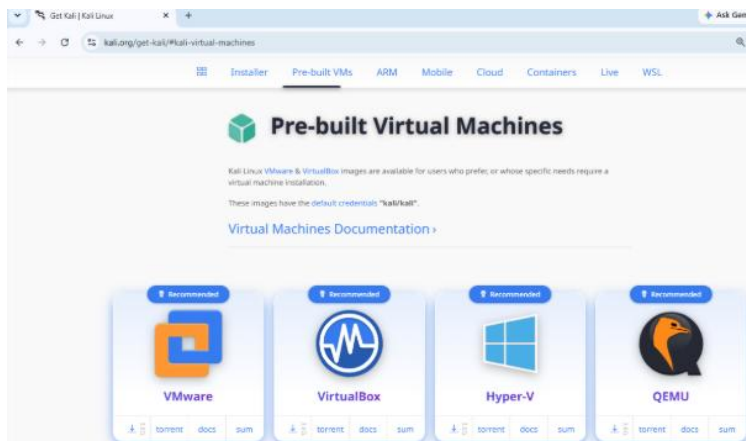
Typically, you would **install VirtualBox → download a Linux ISO (such as Ubuntu) → create a new VM → allocate RAM/storage → attach the ISO → install Linux → install GCC → compile and run C programs.**

2. Installation of VMM & VM

Step-1: Download Oracle Virtual Box for “Windows hosts”

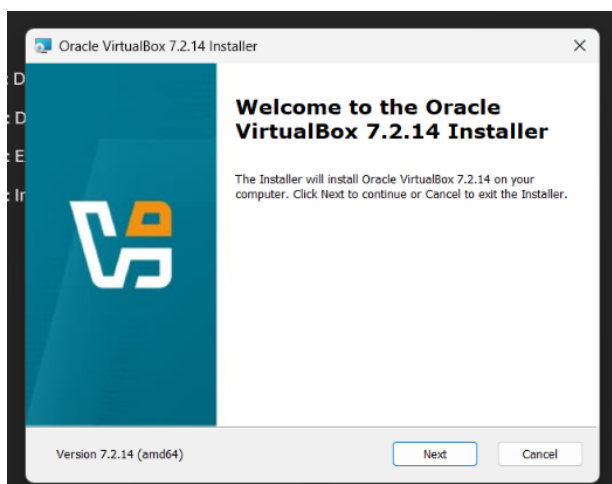


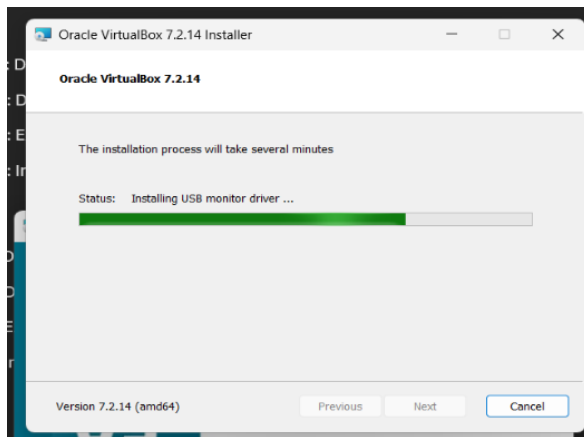
Step-2: Download Pre-built Kali Linux VM for Virtualbox



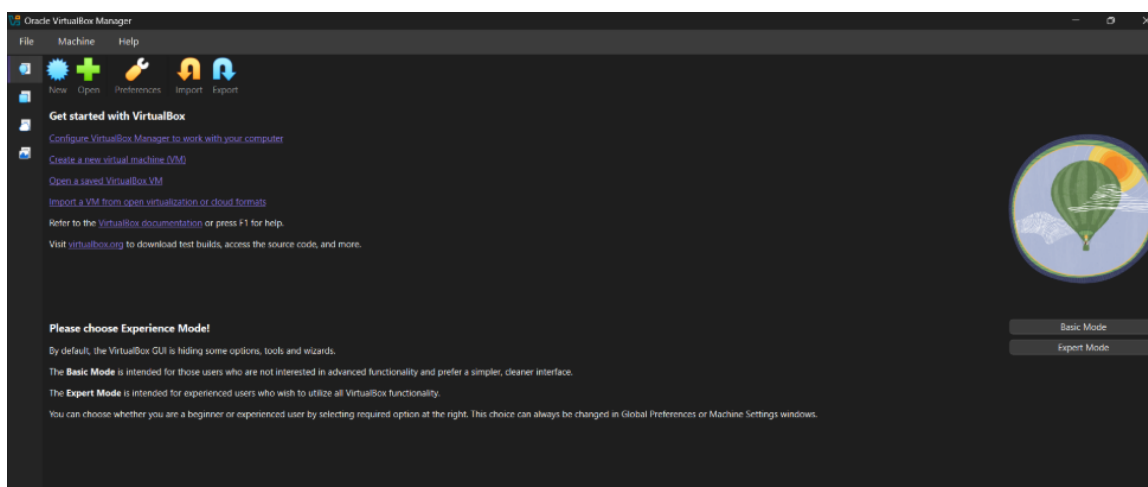
Step-3: Extracted the Kali Linux downloaded file

Step-4: Install Virtualbox

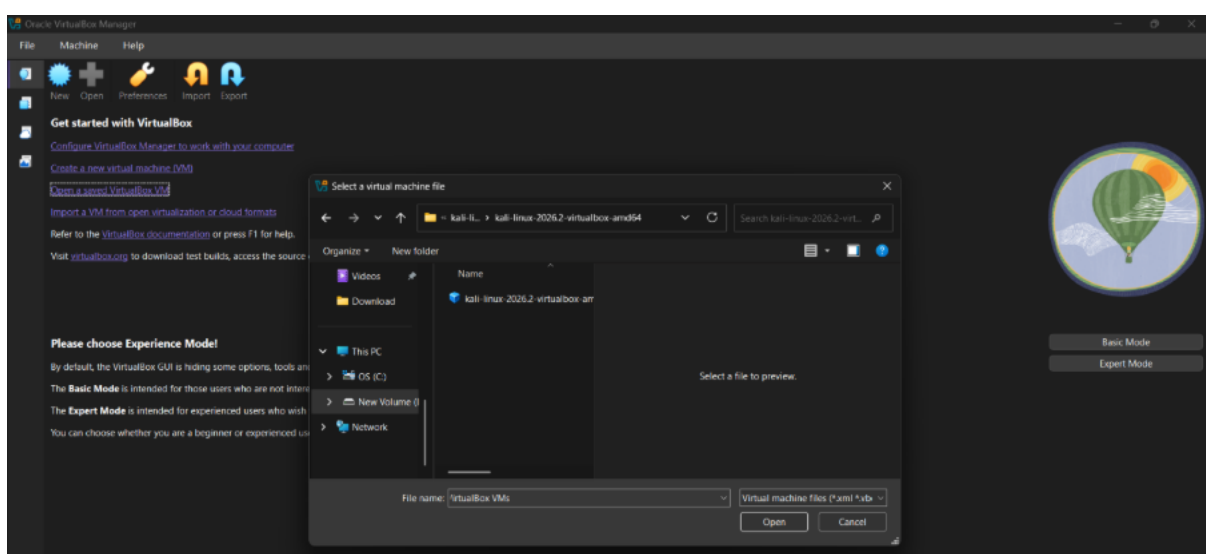


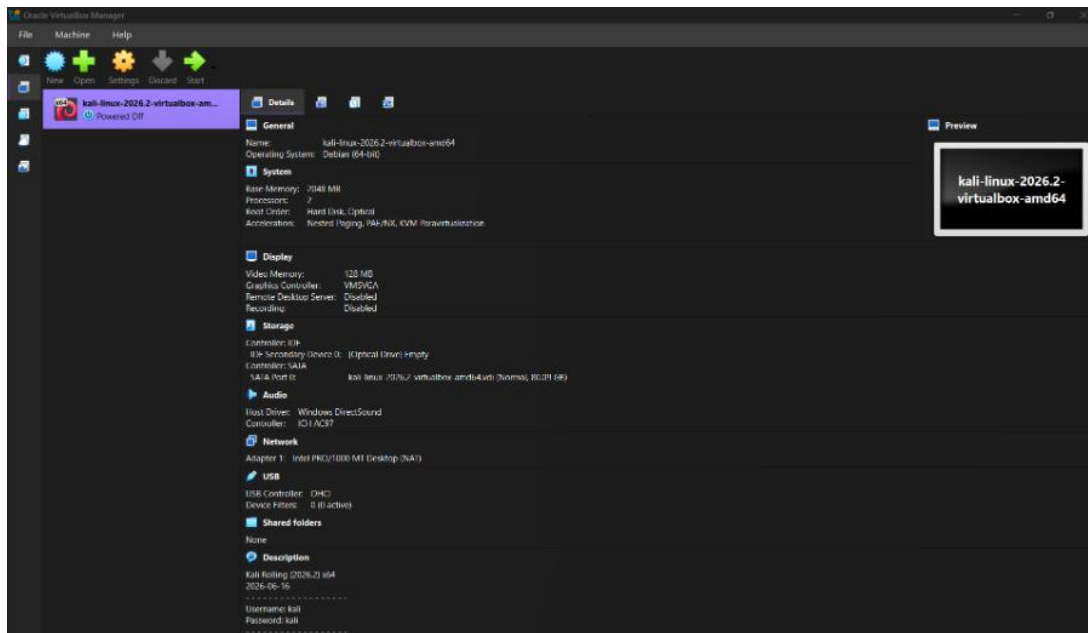


Step-5: Open Virtual box

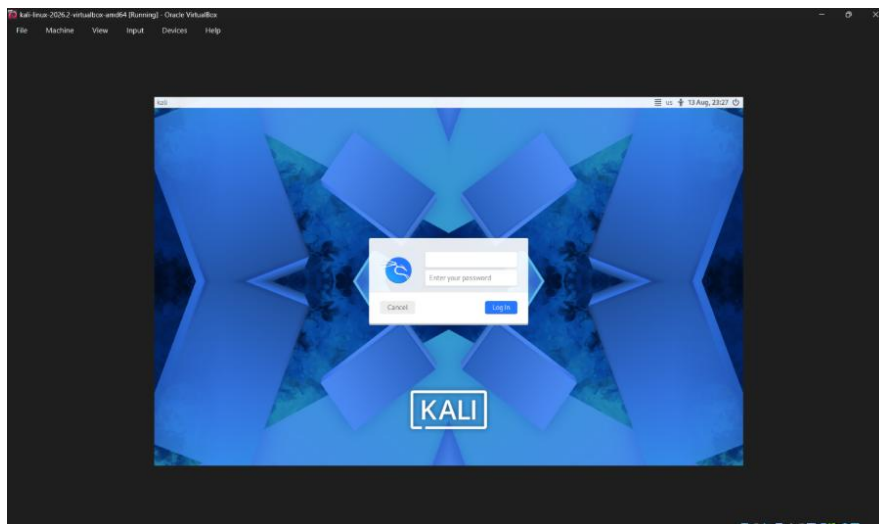
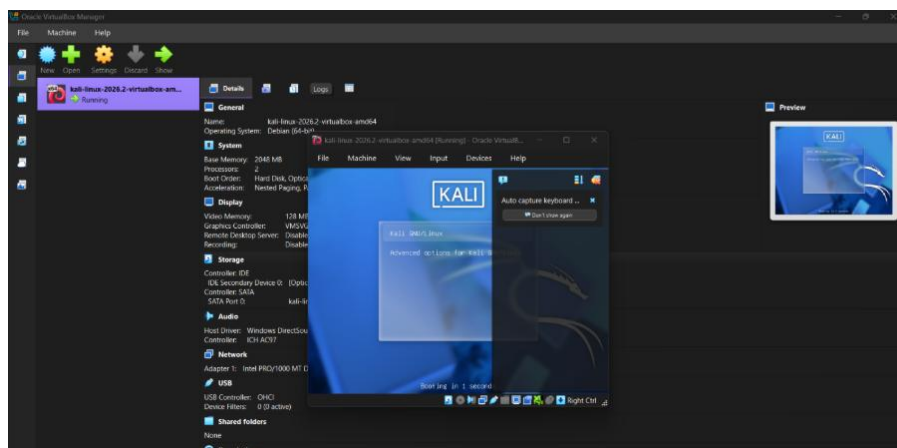


Step-6: Open the “saved VM”





Step-7: Start the VM



Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.

Step-8: Open terminal and create the following simple program, compile and execute to understand the working of the environment.

B) Working with Linux Commands

1. Terminal Basics

Command Brief Explanation

Pwd	Displays the current working directory.
Mkdir	Creates a new directory.
Rmdir	Removes an empty directory.
Dir	Lists files and directories.
Ls	Lists files and directories.
Cd	Changes the current directory.
cd ..	Moves to the parent directory.
Clear	Clears the terminal screen.
History	Displays previously executed commands.
Man	Displays the manual/help for a command.
whoami	Displays the current username.

2. File Commands

Command Brief Explanation

Vi	Opens or edits a text file.
File	Displays the type of a file.
Rm	Removes a file or directory.
Cp	Copies files or directories.
Mv	Moves or renames files/directories.
Cat	Displays the contents of a file.

Command Brief Explanation

More	Displays file contents page by page.
Less	Views file contents page by page with navigation.
Head	Displays the beginning of a file.
Tail	Displays the end of a file.
Chmod	Changes file or directory permissions.

3. Redirection

Operator Brief Explanation

>	Redirects output to a file and overwrites it.
>>	Redirects output to a file and appends to it.
<	Takes input from a file instead of the keyboard.

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